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Research paper

Multi input–Multi output 3D CNN for dementia severity assessment with incomplete multimodal data

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ABSTRACT

Alzheimer's Disease is the most common cause of dementia, whose progression spans in different stages, from very mild cognitive impairment to mild and severe conditions. In clinical trials, Magnetic Resonance Imaging (MRI) and Positron Emission Tomography (PET) are mostly used for the early diagnosis of neurodegenerative disorders since they provide volumetric and metabolic function information of the brain, respectively. In recent years, Deep Learning (DL) has been employed in medical imaging with promising results. Moreover, the use of the deep neural networks, especially Convolutional Neural Networks (CNNs), has also enabled the development of DL-based solutions in domains characterized by the need of leveraging information coming from multiple data sources, raising the Multimodal Deep Learning (MDL). In this paper, we conduct a systematic analysis of MDL approaches for dementia severity assessment exploiting MRI and PET scans. We propose a Multi Input–Multi Output 3D CNN whose training iterations change according to the characteristic of the input as it is able to handle incomplete acquisitions, in which one image modality is missed. Experiments performed on OASIS-3 dataset show the satisfactory results of the implemented network, which outperforms approaches exploiting both single image modality and different MDL fusion techniques.

1. Introduction

Dementia disease consists in the loss of cognitive functioning (thinking, remembering, and reasoning), which interferes with daily life and activities of a person. It describes a syndrome of generalized mental deterioration that also causes personality change and makes the subject vulnerable in many ways (i.e., injuries from falls and accidents) [1]. Alzheimer's disease (AD) is the most common cause of dementia, affecting millions of elderly people around the world. The progression of the disease spans in different stages, from very mild cognitive impairment, where subjects experience memory loss or cognitive difficulties, to mild and severe stages, where damage occurs in the areas of the brain that control languages and conscious thought.

AD is a neurodegenerative disorder, and early detection is a key element to improve the quality of life of affected patients and their families [2]. The Clinical Dementia Rating (CDR) [3] is used in clinical trials to classify the severity of AD. It is derived from an interview with the patient and rates impairment in each of the six cognitive categories (memory, orientation, judgment, community affairs, hobbies, and personal care). The CDR value consists of a five-point scale in which 0 means a cognitive normal condition, 0.5 indicates a questionable impairment, whilst the values 1, 2, and 3 correspond to a mild, moderate, and severe impairment respectively.

In clinical trials, the Magnetic Resonance Imaging (MRI) is the standard diagnostic tool [4,5] due to the fact that the acquired images have a strong relationship with the topology of the brain showing the alteration of the morphology. In particular, the T1-weighted (T1-w) MRI provides information about the brain structure, making it possible to evaluate its volumetric characteristics and atrophy. Positron Emission Tomography (PET) is another imaging technique that consists of the injection of a tracer capable of revealing the metabolic functions of the tissue under analysis. AD is characterized by the presence of β-amyloid ($A\beta$) plaques that may develop many years before the onset of

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dementia [6]. As a consequence, amyloid imaging is widely used in the early diagnosis of AD. The Pittsburgh Compound B (C-PiB) [7] is the first PET tracer specific for $A\beta$ plaques and it is used to identify plaques in brain tissue.

In the last years, Deep Learning (DL) approaches have been employed in medical imaging achieving promising results to support medical decisions [8]. A key role has been played by Convolutional Neural Networks (CNNs), which are networks consisting of several convolutional layers that autonomously learn the set of features that well fits the specific task to solve. Moreover, how to exploit data gathered from heterogeneous sources, such as MRI and PET in the case of AD diagnosis, is an open issue for artificial intelligence research in healthcare, which can be addressed by Multimodal Leaning [9]. It integrates complementary information from diverse modalities of the same phenomena to provide a richer representation than Unimodal Learning. When investigated in conjunction with deep networks, multimodal learning is also known as Multimodal Deep Learning (MDL) [10]. Techniques for multimodal data fusion can be categorized into early, intermediate, and late fusion [10]. Early fusion integrates multiple sources of data into a single features vector, before being used as input to a learning model. Late fusion, also referred as decision-level fusion, combines according to a given rule the decisions from multiple classifiers, each trained on separate modalities. Intermediate fusion, also named as joint learning, exploits the deep neural networks to transform raw inputs into higher-level and shared representations, which are constructed, for instance, by merging into a single layer, units coming from multiple modality-specific paths [10].

There is the need of combining information from heterogeneous and complementary sources, such as MRI and PET, to evaluate the structural and metabolic characteristics of the brain. This makes the MDL approaches well suited in the case of dementia assessment. Although the majority of papers exploit a single modality, that is the MRI, few work have been proposed to provide a model able to process multiple data sources considering the joint learning.

However, when working with a multimodal dataset in the medical field, it is not easy to have images of all the involved modalities, belonging to the same patient. For each subject, *paired acquisition* consists of images coming from all the different sources and collected at the same time or in a specific range. In a real scenario, patients may have *incomplete acquisitions*, in which some modalities are missed. In the literature, some implemented methodologies discard the incomplete instances, considering only the paired acquisitions and limiting the amount of data to be considered [11–13]. On the other hand, few work propose to fill the missing modalities with black images [11] or interpolation operations [14].

In this paper, we conduct a systematic analysis of early, late and joint approaches in fusion for dementia severity assessment on the publicly available OASIS-3 dataset [15], which is the latest release in the Open Access Series of Imaging Studies (OASIS) and includes two different image modalities, the T1-w MRI and the C-PiB PET. We focus on 3D CNN to exploit the volumetric features of the involved images, including in the training step strategies to handle a high imbalance and incomplete dataset. In particular, we analyze the effects of the incomplete dataset in each multimodal fusion technique, and in the case of intermediate fusion, we propose a Multiple Input–Multi Output 3D CNN whose training iteration changes according to the characteristics of the input volumes. The contributions of the proposed methodology can be summarized as follows:

- We evaluate the role of MDL in dementia severity assessment proposing a DL-based approach that exploits the three-dimensional characteristics of the involved medical images.
- We introduce an efficient training strategy to address the challenges posed by an unbalanced dataset with incomplete acquisitions.
- We design and implement a Multiple Input–Multi Output 3D CNN to process both brain MRI and PET acquisitions.

The rest of the paper is organized as follows: Section 2 briefly analyzes the literature, whilst Section 3 introduces the dataset. Section 4 illustrates the implemented methodology, Section 5 describes the experimental setup and Section 6 reports the obtained results. Finally, Section 7 provides some concluding remarks.

2. Related works and motivations

Most of the work in the literature propose solutions that take advantage of DL approaches and MRI for the assessment of dementia. The methodologies implemented differ for the classification task, for the dataset used and for the characteristics of the neural networks (3D CNN or 2D CNN) [16,17]. With reference to the classification task, some work address task distinguishing a normal brain condition from a damaged one [18–22], whilst others deal with a solution for a multiclass classification, also introducing the condition of mild cognitive impairment [23–29]. With reference to the dataset, most of the papers use the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset [30], whereas few work use the OASIS-3 collection [15] to exploit the fine-tuning of state-of-the-art 2D-CNNs (AlexNet [31], ResNet [32], Inception [33], VGG [34]) [18,19,21,23,25,28]. With reference to the neural network characteristics, since MRI is a 3D acquisition, the involvement of state-of-the-art networks introduces the problem of selecting the right number and portion of slices to use. Some articles use the median slice or select some central slices [21,23]. In these cases, each slice is classified independently from the others, and therefore, volumetric information is not exploited.

To overcome this limitation, the 3D CNNs are then introduced permitting to extract the features from the whole 3D brain volume [22,26,27]. However, they require more training time since they are characterized by a huge number of parameters and it is not easy to find pre-trained 3D networks, especially considering biomedical images. The implemented 3D networks in the literature consist of a sequence of 3D convolutional layers, separated by pooling operation, batch normalization, and activation functions.

In the literature, few solutions considering PET acquisitions have been proposed, presenting the same heterogeneity as seen in work focusing on MRI. Some papers focus on the ADNI dataset [30], exploiting fluorodeoxyglucose (FDG) PET. In particular, the solutions presented in [35,36] use state-of-the-art CNNs (AlexNet [31] and InceptionV3 [33]), while the methodology proposed in [37] relies on a 3D CNN. Only one paper [38] uses the OASIS-3 dataset [15], proposing a 3D CNN trained from scratch, focussing on amyloid imaging.

In the recent years, implemented solutions exploited joint learning for dementia assessment, without considering the late and early fusion. The authors of [11] proposed a methodology that combines T1-w MRI, Diffusion Tensor Imaging (DTI-MRI) and 2D ResNet18 [32], representing the first article that focuses on a multimodal approach with the OASIS-3 dataset [15].

The majority of papers that use both MRI and PET images focus on ADNI dataset [30]. In [14] authors used a 2D CNN for the classification of features extracted from MRI, PET and genetic data, while in [12] authors proposed a multi-input 3D CNN considering as input the hippocampal area selected from MRI and PET. In [39] sparse autoencoder and 3D CNN were introduced for classification, while in [13] stacked Deep Polynomial Networks (DPNs) were used to extract features from the two image modalities. The solutions proposed in [12–14,39] report more than 90% of accuracy in separating a normal brain condition from a damaged one.

As aforementioned in Section 1, when working with a multimodal environment, it is not easy to have images of all the involved modalities, that, in turn, will result in the need to manage *incomplete acquisitions*. Authors in [12,13] used only a paired dataset, reducing the number of images available, while the solution proposed in [11] replaced the missing modality with black images. In [14] the authors used linear interpolation to fill in the missing modalities. However, this

process makes the different acquisitions dependent on each other, especially in the case of longitudinal studies. Authors in [40] handled the incomplete dataset proposing a fully connected multi-input network, consisting of three fully connected layers, with three outputs (first modality only, second modality only, paired datasets). The features are extracted from MRI and PET before being fed into the network that is updated in different parts according to the available input data, obtaining an accuracy equal to 65%.

As a further remark, we would point out that the OASIS-3 [15] and ADNI [30] datasets consist of longitudinal studies, so that they include images of the same patient collected at different time. This implies that a time-dependent relationship between images belonging to the same subject exists and has to be taken into account, especially when designing the validation step of the implemented solutions. However, although some work in the literature got very high precision (more than 90%), most of them conducted an image level rather than a patient-level split of the dataset [11,13,14,18,19,21,23,26–29,37,39], not guaranteeing complete independence between the train, validation, and test sets, as images belonging to the same patient may not be included in the same set. Moreover, some work applied data augmentation techniques before splitting the dataset [19,26–28]. We argue that these characteristics may introduce bias in the performance evaluation step.

In this paper, we focus only on the OASIS-3 [15] dataset, using the CDR value to provide an assessment of the dementia status rather than a specific diagnosis. Indeed, as reported in [15], the diagnostic information is separated from the clinical assessment that includes the CDR value. It is worth noting that in the ADNI [30] dataset the information about the CDR value is not available for all the subjects, since the study focuses on the progression of mild cognitive impairment and early AD, presenting a specific diagnosis for each image. This makes the dataset suited for tasks that aim to determine a diagnosis or a specific Alzheimer's stage rather than an assessment of dementia status.

In comparison with the existing literature, our work introduces several key innovative aspects:

- To perform a systematic analysis of MDL approaches for dementia status assessment considering both MRI and PET images. To this end, we exploit different fusion techniques such as early, late, and intermediate fusion.
- To implement a solution completely based on 3D CNN, extracting features from the whole 3D brain volume.
- To propose a training strategy capable of handling a highly imbalanced and incomplete dataset. Notably, in the case of the intermediate approach, we implement a Multi Input–Multi Output 3D CNN architecture that adapts its training iterations according to the input characteristics.

3. Population

In this work, we consider the OASIS-3 dataset [15], consisting in a compilation of MRI, PET imaging, and clinical data for 1098 patients. The study collects 605 cognitively normal patients and 493 individuals at various stages of cognitive decline. All the MRI, PET, and clinical data acquisition sessions report information about when they are acquired expressed as the number of days since the subject's entry date into the study. The status of dementia was assessed using the Clinical Dementia Rating Scale (CDR). For each assessment, information on the CDR value is available, together with a diagnosis of dementia (i.e., "cognitively normal", "AD dementia", "vascular dementia"). We focus only on the most common cause of dementia, Alzheimer's disease, selecting among subjects with CDR> 0, individuals with AD diagnosis, reducing the number of patients to 1025. In this work, we exploit T1-w MRI and C-PiB PET acquisitions, since they are the most used diagnostic tools for dementia diagnosis and assessment [4,5,7].

The CDR value is used to distinguish between three different classes, as proposed in [18,23,24,41,42]: the CDR value set at 0 ($CDR = 0$) represents a normal cognitive function, determining the cognitive normal class (C); the CDR value set at 0.5 ($CDR = 0.5$) indicates very mild dementia, determining the mild class (M); the CDR value greater than 0.5 ($CDR > 0.5$), represents mild/severe dementia, suggesting the severe class (S). Hence, the task to solve consists in the classification of each MRI, PET, and MRI-PET pair into three different classes (C, M, and S), which we address considering two different approaches:

- **Three-class classification (Task A)**: it directly classifies samples into one of the three different classes (C, M, S).
- **Hierarchical classification (Task B)**: it consists of two steps:
  - *Cognitive Normal Classification (Task B1)*: which distinguishes between healthy (C) and diseased (NC) instances.
  - *Disease Classification (Task B2)*: which discriminates between M and S that is applied if the instance is classified as NC.

Scan sessions and clinical data acquisition do not always occur at the same assessment. As a consequence, there is the need to find a criteria to match up MRI, PET images, and clinical data, with the aim of attributing the CDR value to each MRI and PET acquisition. A common criterion is to consider all clinical information within a year before and after the date of the MRI (or PET) session valid. In particular, we consider the clinical data closest to each MRI (or PET) session to be a match if the difference in days is less than one year. We match the MRI and PET scans to clinical assessments separately, and the result consists of three different datasets, namely MRI, PET and PAIRED datasets. The MRI and PET datasets include all the MRI and PET volumes, respectively, with the related CDR value, while the PAIRED dataset contains only MRI-PET pairs, represented by the MRI and PET volumes associated with the same clinical information date. Straightforwardly, both the MRI and PET datasets include the acquisitions of the PAIRED dataset. It is worth noting that the set of data involved in this paper represents an incomplete dataset in which, for each assessment, all image modalities are not always available.

During the data analysis step, 46 patients were deleted because the pre-processing phase of the raw acquisitions failed, as reported by the authors in [15]. Moreover, Fig. 1 shows the case where MRI or PET images are not available for each clinical assessment. In particular, the patient OAS31045 should move from a normal condition to a mild/severe dementia according to the information coming from clinical data. However, only volumes representing the normal condition are provided. This characteristic implies that we should consider the available volumes when studying the distribution of patients in the different classes. We discard all the patients that according to the clinical information change their condition from CDR = 0 (C class) to CDR > 0 (M and S classes), but only images for C are available. As a consequence 100 subjects are deleted. Then, with the aim of maintaining the same age distribution in the three classes, we discard patients younger than 50 years. In the *Supplementary Material* we provide all the details about the preparation of the dataset, including the rules implemented to filter different subjects.

In this study, we consider a total of 628 patients, including 1263 MRI and 711 PET acquisitions, representing the MRI and PET datasets respectively, while the number of MRI-PET pairs is 628. Table 1 shows the progression of the patients in the three classes. The small number of patients in class S does not allow a further division of the CDR value. Table 2 reports the information on the generated datasets considering the number of volumes for each class. It is worth noting that the MRI and PET datasets include the volumes in the PAIRED one.

Table 1
The progression of patients in the three classes, cognitive normal (C), very mild dementia (M), mild/severe dementia (S).

Start with	End with			Total
	C	M	S	
C	340	67	3	410
M		145	10	155
S			63	63
Total	340	212	76	628

Table 2
Information about generated datasets in terms of number of volumes for each class, cognitive normal (C), very mild dementia (M), mild/severe dementia (S).

Dataset	Classes			TOTAL
	C	M	S	
MRI	926	258	79	1263
PET	627	75	16	711
Paired	556	58	14	628

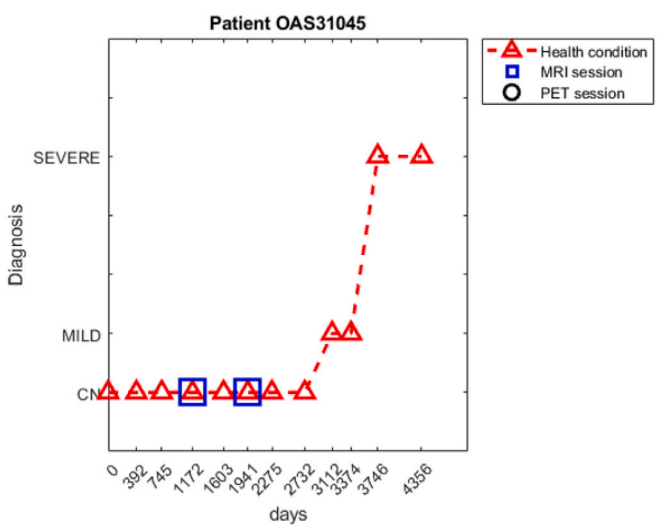


Fig. 1. Disease progression for patient OAS31045.

4. Methodology

This work aims to develop a decision system supporting dementia severity assessment, and exploiting information coming from two different image modalities, namely the T1-w MRI and the C-PiB PET. To this end, we investigate different MDL approaches belonging to all the three paradigms (early, late and joint). As described in Section 3, we address the distinction across three classes considering two different approaches, namely task A and B. Hierarchical classification is introduced to explicitly exploit the relationship between classes related to different CDR levels (C is $CDR = 0$, M is $CDR = 0.5$ and S is $CDR > 0.5$). Moreover, we want to analyze the differences with the three-class classification approach, where the three classes are considered as independent, delegating the model the task of learning their relationships.

The proposed methodology consists of two steps: *Pre-processing*, used to prepare MRI and PET, and the *Dementia Severity Assessment*, which introduces the implemented solutions considering multimodal approaches.

4.1. Pre-processing

T1w-MRI volumes are processed according to the procedure reported in [15], including motion correction, removal of non-brain tissue

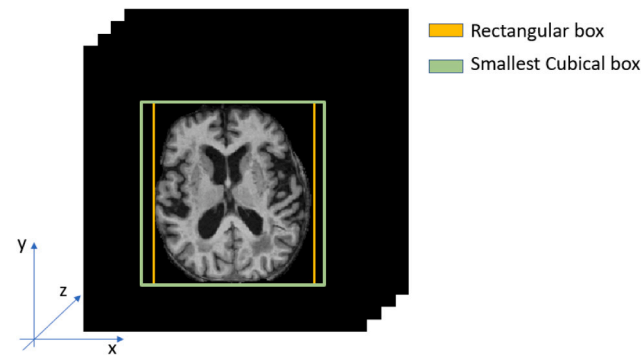


Fig. 2. Extraction of brain region for each patient: the smallest 3D cubical box (in green) is computed from the rectangular box (in orange) surrounding the brain. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(skull-stripping) and intensity normalization. The result is a set of volumes of size 256 × 256 × 256 with isotropic dimension.

A PET scan consists in the acquisition of 3D volumes after the injection of the tracer, resulting in 4D data. In OASIS dataset subjects underwent a 60 min dynamic PET scan in 3D mode (24 × 5 s frames; 9 × 20 s frames; 10 × 1 min frames; 9 × 5 min frames) generating a maximum of 52 3D volumes acquired over time. Since we are interested in C-PiB retention, only volumes acquired after tracer absorption are considered. Indeed, the acquisitions obtained 40 min after tracer injection [43] are averaged to create a *static* PET scan, which is then rigidly registered to the MRI volume acquired during the dynamic PET scan session using mutual information as a similarity metric [44,45]. The result is a PET volume of size 256 × 256 × 256 with isotropic dimension. To reduce the amount of non-brain tissue to include, the smallest 3D cubical box including the patient's brain is extracted from each MRI and PET volume. The cubical box is patient-dependent and its dimension strongly depends on the y axis, as shown in Fig. 2 . The resulting volumes are then normalized in [0,1] to ensure that, in the next stage, the involved CNNs operate on images of the same scale across different acquisitions.

4.2. Dementia severity assessment

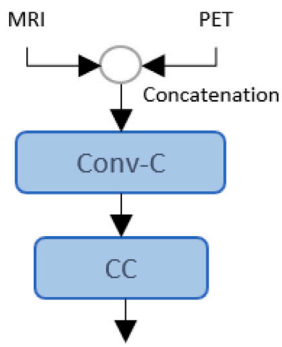
In clinical trials, T1-w MRI and PET are the standard diagnostic tools used to assess the severity of dementia. They provide different information since the former focuses on the volumetric and structural characteristics of the brain, while the latter reveals its metabolic functions. MDL allows the fusion of complementary information coming from heterogeneous sources with the aim of providing a richer data representation than the unimodal approach.

Among all deep neural networks, CNNs are widely used in biomedical image processing [8,46] with surprising results. A typical CNN consists of stacked *relatively complex layers* [47], with each of them having a convolutional stage, a non-linearity function (i.e ReLU), and a pooling operation. The set of *complex layers* constitutes the Convolutional Core (Conv-C), responsible for the features extraction step, while the classification is performed by a Classification Core (CC), typically including fully connected layers.

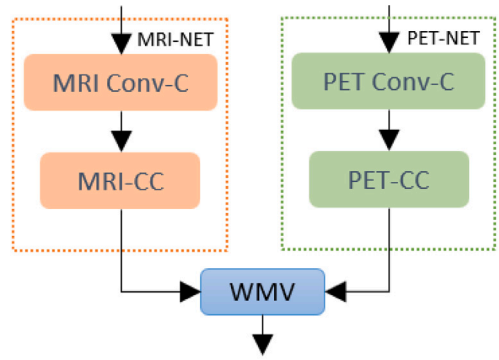
We exploit CNNs to process MRI and PET scans for the dementia severity assessment. Each acquisition represents a 3D volume, with three spatial dimensions (x, y, z) resulting in the need of using networks with 3D convolutional layers. Information coming from complementary data sources is exploited considering MDL techniques. Fig. 3 illustrates the fusion approaches in multimodal learning.

In the early fusion (EF) approach, described in Fig. 3(a), each pair of MRI-PET is organized in a single structure before being considered as an input for a CNN. To this end, we convert each MRI-PET

(a) Architecture of the 3D CNN proposed for the EF



(b) Architecture of the 3D CNN proposed for the LF



(c) Architecture of the 3D CNN proposed for the IF: it is possible to distinguish three different sub-networks, the MRI-NET, the PET-NET, and the PAIRED-NET.

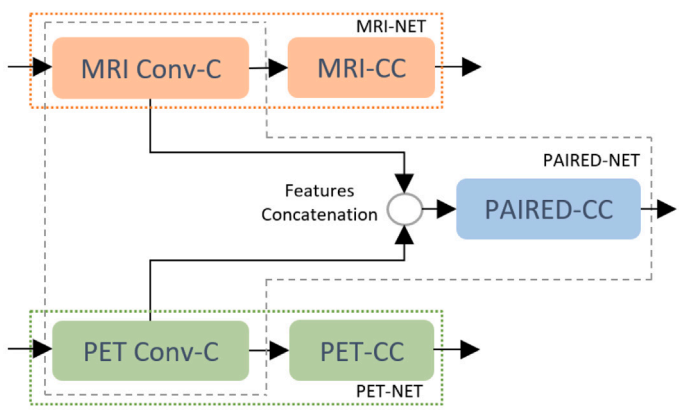


Fig. 3. Fusion approaches in MDL for dementia assessment: 3(a) EF, 3(b) LF, 3(c) IF.

pair into a 2-channel volume by concatenating the different paired acquisitions. Straightforwardly, we change the first layer of the 3D-CNN, by modifying the number of the input channels. In EF approach, two image modalities simultaneously contribute to prediction, using different information at the same time. Furthermore, we train the network using the PAIRED dataset, which has a reduced amount of available data.

The late fusion (LF) completely relies on the unimodal approach since it aggregates the prediction coming from the classifiers implemented for the MRI and the PET separately, as shown in Fig. 3(b). To this goal, we combined the predictions coming from the two networks using the Weighted Majority Voting (WMV), in which a weight is assigned to each prediction according to the network output probability. However, in LF each CNN acts independently, not taking advantage of the complementary characteristics of the two image modalities that do not influence each other during the prediction.

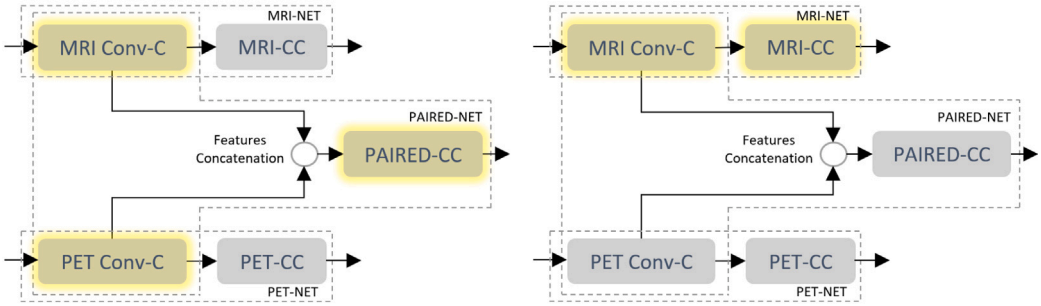
In the case of intermediate or joint fusion (IF), we propose a Multi Input-Multi Output 3D-CNN architecture, that simultaneously processes MRI and PET scans. Furthermore, we introduce a training strategy able to handle the case where one of the inputs of the two modalities is missing. This permits us to avoid reducing the training dataset, thus exploiting all the available volumes. Fig. 3(c) shows the architecture of the proposed network, which consists of two inputs, one for the MRI and one for the PET volumes, respectively; then there are two convolutional cores, namely the MRI Conv-C and the PET Conv-C, which extract abstract representations that are the inputs of two classification cores, namely the MRI-CC and the PET-CC. Next, the joint fusion concatenates the outputs of the two convolutional cores, which fed a PAIRED classification core, named as PAIRED-CC (Fig. 3(c)). Therefore, the result is a 3D-CNN with three outputs, with only one of them being considered according to the nature of the input: the PAIRED-CC is considered only for *paired acquisitions*, while the MRI-CC and the PET-CC act in the case of incomplete inputs. In particular, it is worth noting that we can distinguish three different sub-networks, namely the MRI-NET, the PET-NET, and the PAIRED-NET that, during the training, are separately updated as follows (Fig. 4):

1. When both modalities are available, the volumes of each MRI-PET pair fed MRI Conv-C and PET Conv-C, respectively, whose outputs are concatenated and considered as input for the PAIRED-CC for the classification. The described chain represents the PAIRED-NET, whose classification loss is used to update the two convolutional cores and the PAIRED-CC element (Fig. 4(a)).
2. When the PET data is missing, the MRI volumes fed into the MRI-NET, consisting of MRI Conv-C and MRI-CC. The output of the MRI-CC is used to compute the classification loss, updating the network weights (Fig. 4(b)).
3. Similarly when the MRI data is missing, the PET volumes fed PET-NET consisting of the PET Conv-C and PET-CC chain. The output of the PET-CC is used to compute the classification loss, updating the network weights (Fig. 4(c)).

It is worth noting that the joint fusion approach described so far overcomes the aforementioned disadvantages of both the early and the late fusions. Indeed, on the one hand, it allows to train the network

(a) In the first iteration, the PAIRED-NET is considered updating the parts highlighted in yellow.

(b) In the second iteration, the MRI-NET (highlighted in yellow) is updated since the input is a set of MRI volumes



(c) In the third iteration, the PET-NET is considered updating the parts highlighted in yellow.

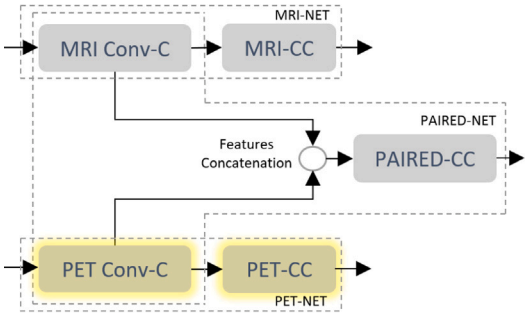


Fig. 4. Implemented training strategy to handle an incomplete dataset.

exploiting all the data available in the dataset. On the other hand, it is also beneficial in the test step, as it is possible to classify instances where one modality is missing.

Furthermore, if we remove the PAIRED-CC, we can consider the MRI-NET and the PET-NET as independent networks, obtaining the unimodal approach (U). In view of this, we denote U MRI-NET and U PET-NET as the networks trained with the MRI and PET volumes respectively.

4.2.1. CNN architecture

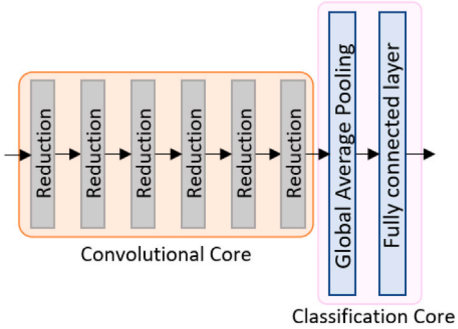
In this subsection, we are presenting two different deep models we implement for the convolutional part (Conv-C) and for the classification part (CC) of each architecture, regardless of the input or the type of fusion adopted, i.e., early, late and joint. These two networks are named as BasicNet and ResNet, and when presenting each of them we will first describe Conv-C and then we will focus on CC.

BasicNet consists of six *reduction layers* followed by a global average pooling and a fully connected layer. A reduction layer is a block with a 3D-convolutional operation consisting of $3 \times 3 \times 3$ kernel, stride, and padding set to 2 and 1 respectively, followed by batch normalization and ReLU function. Each convolution reduces the input feature map and doubles the number of channels, excluding the first convolutional layer with 16 output channels. The stride set to 2 makes pooling operations (max or average pooling) not necessary [48]. The chain of the six reduction layers constitutes the Conv-C of the network, which is responsible for the features extraction step, while the global average pooling and the fully connected layer represent the CC as shown in Fig. 5(a). In the implemented methodology BasicNet is trained from scratch.

Inspired by the state-of-art network proposed in [32], ResNet is a 3D CNN, adapted to handle 3D volume data. Furthermore, the solution implemented in [49] offers a set of 3D ResNet architectures pre-trained with medical images for segmentation tasks. The architecture consists of a backbone, that is an encoder representing the ResNet basic structure, responsible for features extraction, and a set of decoders for the generation of the segmentation masks. Hence, we adapted the ResNet for the classification task as proposed in [49]: we retain the backbone while replacing the set of decoders with a global average pooling and a fully connected layer, which represent the CC. Fig. 5(b) shows the resulting ResNet architecture highlighting the backbone, that is the Conv-C, consisting of a first convolutional layer, followed by batch normalization, ReLU and Max Pooling layers, and a chain on four layers containing the blocks implementing the residual network as proposed in [32]. Different from [49], we restore the stride of the convolution kernels in blocks 3 and 4 equal to 2 allowing down-sampling the features maps to better tailor the network to a classification problem. The result is an architecture similar to the one proposed in [32], with some modifications: we use 3D convolutional layers to process MRI and PET acquisitions, and the input is a 1-channel 3D image. Note that on the one hand, we used transfer learning, considering the pre-trained backbone as a starting point and implementing the fine-tuning for adapting the ResNet for the specific task to solve. On the other hand, the classification core is trained from scratch.

We test all the techniques described in Section 4.2 with the two 3D-CNNs. In particular, in the U and LF approaches we use the architectures as presented above, while in the EF the first convolutional layer is modified to handle a 2-channel volume. The BasicNet and the ResNet are also exploited in the IF implementing the MRI-NET (MRI Conv-C and MRI-CC) and the PET-NET (PET Conv-C and PET-CC). The PAIRED-NET is constituted by the MRI Conv-C, the PET Conv-C, and the PAIRED-CC, consisting of a global average pooling and two fully connected layers separated by the ReLU function.

(a) BasicNet architecture: the network consists of six *reduction layers*, representing the Convolutional Core (Conv-C), followed by a Global Average Pooling and a fully connected layer, constituting the Classification Core (CC)



(b) ResNet architecture: the Convolutional Core (Conv-C) consists of a first convolutional layer, followed by batch normalization, ReLU and Max Pooling layers, and a chain on four layers containing the blocks implementing the residual network, while the Global Average Pooling and the fully connected layer represent the Classification Core (CC)

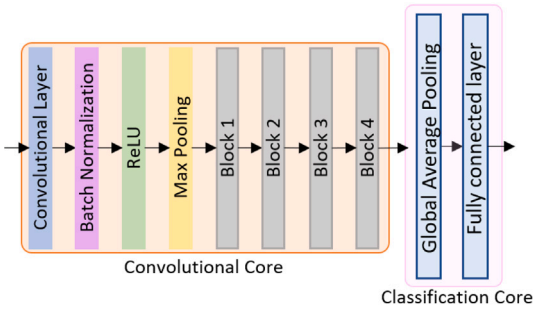


Fig. 5. Architecture of the implemented CNNs.

Let us now turn the attention to the training process. In the medical field, it is very difficult to have large and balanced datasets as the number of patients usually involved in the study is small and the number of acquisitions belonging to healthy subjects is greater than the number of unhealthy ones. In this work, to improve the network's robustness we introduce variability in the set of data used for training, by applying data augmentation techniques which include translation, rotation, and scaling [49]. Straightforwardly, we avoid completely overturning the position of the brain areas when setting up the augmentation operations. Since the extracted bounding box is strongly influenced by the y axis, volumes are translated within $[-10, 10]$ pixels in x and z dimensions, and within $[-5, 5]$ in the remaining one. The rotation angle is selected within $[-10°, 10°]$ for the y axis and within $[-5°, 5°]$ for the remaining ones, to reproduce natural head positions. The scaling factor, applied in each dimension, is chosen within $[0.9, 1.1]$, to simulate different brain sizes by introducing moderate modifications.

During the training, we also implemented a strategy to handle data imbalance, ensuring the creation of balanced batches in the various iterations. Denoting with b the batch size, and with N_c the number of classes, each batch contains b/N_c instances of each class. As a consequence, in each iteration, the network receives a balanced batch. However, since the number of batches to create in an epoch strongly depends on the size of the majority class, different batches may share samples belonging to the minority classes, resulting in an approach similar to that achieved by the simple replication of instances.

5. Experimental set-up

The 3D CNN presented for IF consists of three different sub-networks, the MRI-NET, the PET-NET and the PAIRED-NET, as described in Section 4.2 and Fig. 3(c). As a consequence, it is possible to evaluate the performance of the implemented network on the three datasets, considering the respective classification cores. However, when the input consists in a MRI-PET pair, the three sub-networks can also be used in parallel, resulting in three outputs. To provide a unique prediction, a WMV combining strategy is used. We refer to the described configuration with IF-WMV. Furthermore, the IF approach can be evaluated on a dataset obtained by merging the MRI, PET, and PAIRED datasets (COMPLETE dataset). Indeed, if the input is an MRI (or PET) volume, the output of the MRI-NET (or PET-NET) is considered, while if an MRI-PET pair is considered, the results of the three sub-nets are aggregated using WMV. As a consequence, a performance evaluation can be obtained considering all available volumes.

The EF and LF approaches can be used only with the PAIRED dataset, since the first requires an input of two channels, while the latter combines the results of two networks. However, it is possible to evaluate the performance by considering the COMPLETE dataset exploiting the network trained in the U approach for incomplete inputs. More in detail, in the LF the output combination is performed only for the MRI-PET pairs, while in the EF the outputs coming from the three CNNs are aggregated using WMV in the case of paired input.

We provide a complete set of experiments since we exploit the MDL fusion methods in each task, namely the three classes (Task A) and hierarchical (Task B) classification, also detailing the results on the Tasks B1 (C/NC classification) and B2 (M/S classification). Moreover, we test the EF, LF and IF techniques with the two CNNs, BasicNet and ResNet.

To assess the effectiveness of the proposed methodology, we compare the result of the IF, EF and LF with the U approach, considering also a proposal from the literature focusing on the OASIS-3 dataset [15].

The U approach is used as a baseline, evaluating its performance on the MRI and PET datasets. In particular, we train the BasicNet and the ResNet considering the MRI and PET volumes separately, obtaining the U MRI-NET and the U PET-NET.

We then select the solution described in [11] as it represents, to the best of our knowledge, the first article that focuses on MDL considering the OASIS-3 dataset [15]. Since the authors use T1-w and DTI MRI, we carefully implemented the proposed methodology according to the information provided in [11], exploiting T1-w MRI and C-PiB PET for the classification of the three classes (Task A). Indeed, Massalimova et al. [11] explored the fine-tuning of the state-of-the-art RestNet18 [32] in the U approach, proposing in the IF a multi-input 2D CNN that leverages the network trained in the solution with a single modality. The input consists of a 3-channel image obtained concatenating the median slices of the sagittal, axial, and coronal views from each 3D scan. In the IF approach, the authors replace the missing modalities with black images. We compare our methodology with the solution proposed in [11] implementing both the U and IF approaches. Table 3 gives details in terms of datasets and tasks for each of the considered approaches.

We organize the experiments in four different groups, as summarized as follows:

- Experiment 1 (EXP.1): we focus on Task B1 (C/NC classification), evaluating the effectiveness of the MDL and U approaches on the four datasets (MRI, PET, PAIRED, COMPLETE).
- Experiment 2 (EXP.2): we consider Task B2 (M/S classification), exploring both the MDL and U solutions.
- Experiment 3 (EXP.3): we focus on Task A (C/M/S classification), evaluating the effectiveness of the MDL and U approaches on the four datasets (MRI, PET, PAIRED, COMPLETE).
- Experiment 4 (EXP.4): similarly, we focus on Task B (hierarchical classification.

As described in Section 4.1, the smallest 3D cubical box including the patient's brain is extracted from each MRI and PET volume. Since the resulting volume in patient-dependent, a resize stage is used before feeding data into the involved 3D CNNs. More in detail, the BasicNet and ResNet receive as input volumes of size $128 \times 128 \times 128$. In the IF technique, fine-tuning is exploited by initializing the MRI-NET and the PET-NET with the weights determined in the U approach, while the PAIRED-CC is trained from scratch. In experiments involving ResNet, we use the ResNet34 architecture. The involved networks are trained by minimizing the cross-entropy loss. The maximum number of epochs is 200, the batch size is 32 for BasicNet and 16 for ResNet. For each epoch, we evaluate the loss function on the validation set and then, after the last epoch, we use the best weights (i.e. those that minimize the loss function on validation set) of the network to compute the performance on the test set. The learning rate for the cross-entropy loss was set to 10^{-5} and 10^{-6} for the experiments involving BasicNet and ResNet, respectively. We used Adam optimizer with a weight decay set to 10^{-4}. In the *Supplementary Material* we provide further information about the hyper-parameters used in the experiments.

Performance is evaluated in terms of Accuracy (ACC), Precision, and Recall for each class, and Area under the ROC Curve (AUC). For the three-class classification, the average AUC weighted by each class support, that is the number of true instances for each label, is reported.

Table 3
Details in terms of datasets and tasks for each of the considered approaches.

Approach	Dataset	Task
EF (BasicNet, ResNet)	PAIRED, COMPLETE	B1, B2, A
LF (BasicNet, ResNet)	PAIRED, COMPLETE	B1, B2, A
IF (BasicNet, ResNet)	PAIRED, COMPLETE, MRI, PET	B1, B2, A
U (BasicNet, ResNet)	MRI, PET	B1, B2, A
ResNet18[11]	PAIRED, COMPLETE, MRI, PET	A

Moreover, to take into account the unbalanced nature of the involved dataset, we also report the Balanced Accuracy (BAcc), computed by averaging the recall for each class, and the F1-score, obtained by including each class support in the computation. All experiments were run in a 5-fold cross-validation (CV) setting, to better assess the generalization ability of the implemented approaches. In each iteration (i) of the CV, three folds represent the training set, the ith fold is used as the test set, while the $(i-1)$th one is considered as validation set. More in detail, we perform patient-based cross-validation, avoiding the use of volumes from the same patient during the training and evaluation phase.

All the experiments were carried out using Pytorch (version 1.10), while the pre-processing step was implemented in MATLAB 2020b. We used a Linux workstation equipped with two NVIDIA RTX 3090 GPUs, AMD Ryzen Threadripper 3960X 24-Core Processor, 64 GB of DDR4 RAM. The source code of the implemented experiments is available at this link.[1]

6. Results

The results are organized by each class of experiments (EXP.1, EXP2, EXP3, EXP4) and are detailed for each dataset (MRI, PET, PAIRED, and COMPLETE), considering both networks (BasicNet and ResNet). This approach facilitates a comprehensive performance comparison among different methods. In each task, our objective is to evaluate the performance of the proposed Multi-Input Multi-Output 3D CNN, representing an innovative IF method capable of handling incomplete samples. The existence of three classification cores allows us to present results for all four datasets. Specifically, for MRI and PET sets, we focus on the outputs of MRI-CC and PET-CC separately, comparing them with the U approach, where U MRI-NET and U PET-NET serve as baselines. Furthermore, when analyzing PAIRED and COMPLETE datasets, we extend our comparison to include other implemented multimodal fusion strategies, namely EF and LF.

The tables described in this section report the performance of EXP.1, EXP.2, EXP.3, EXP.4 respectively, specifying the dataset (Data), the network (Net), the fusion modality (Mod.), the accuracy (ACC), Precision, Recall and AUC. We denote as Ba and Re the BasicNet and ResNet architectures. For each metric, we report the average rate computed adopting 5-fold CV and its standard deviation, omitting 0. after the $\pm$ symbol and showing only 2 significant digits. Moreover, we include the BAcc and the F1-score in bar plots to have a summary of the results of each experiment.

For the sake of completeness, we also visualize the training loss curves of the proposed IF methodology. As aforementioned in Section 5, we conducted a 5-fold cross-validation for each experiment, resulting in five curves for both the training and validation sets. To provide a more concise representation for each classification task, we calculated the mean value and standard deviation of the loss function for each epoch. The bold lines represent the average training and validation curves, while the shaded areas indicate the region defined by the standard deviation. In the *Supplementary Material*, we show the training and validation curves for each classification task and fusion modality, including the unimodal approach.

[1] https://github.com/priamus-lab/Multi-Input---Multi-Output-3D-CNN-for-Dementia-Severity-Assessment-with-Incomplete-Multimodal-Data/

Table 4
Performance of the EXP.1 (Task B1) the MRI dataset. Ba and Re denote the BasicNet and ResNet architectures.

Data	Net	Mod.	ACC	Precision		Recall		AUC
				C	NC	C	NC	
MRI	Ba	U	76.96±.02	89.54±.01	55.00±.01	77.65±.02	75.07±.01	79.80±.01
		IF	78.78±.05	89.35±.03	58.08±.06	80.67±.08	73.59±.09	80.67±.03
	Re	U	80.36±.01	91.96±.01	59.78±.02	80.24±.02	80.71±.03	81.71±.02
		IF	**84.80±.01**	**93.28±.01**	**67.47±.05**	**85.42±.02**	**83.09±.03**	**84.30±.02**

Table 5
Performance of the EXP.1 (Task B1) the PET dataset. Ba and Re denote the BasicNet and ResNet architectures.

Data	Net	Mod.	ACC	Precision		Recall		AUC
				C	NC	C	NC	
PET	Ba	U	78.90±.03	94.51±.01	33.88±.06	80.48±.04	68.13±.06	77.86±.05
		IF	80.45±.01	96.34±.01	37.50±.04	80.65±.01	79.12±.04	79.39±.03
	Re	U	88.33±.04	95.90±.02	53.17±.07	90.48±.03	73.63±.13	79.00±.07
		IF	**89.59±.04**	**96.91±.01**	**56.59±.18**	**90.97±.05**	**80.22±.07**	**84.50±.04**

Table 6
Performance of the EXP.1 (Task B1) the PAIRED dataset. Ba and Re denote the BasicNet and ResNet architectures.

Data	Net	Mod.	ACC	Precision		Recall		AUC
				C	NC	C	NC	
PAIRED	Ba	EF	82.96±.03	93.42±.03	34.23±.08	86.87±.05	52.78±.20	78.85±.08
		LF	82.48±.02	96.65±.02	37.33±.07	83.09±.05	77.78±.20	87.37±.70
		IF	90.13±.03	96.25±.01	55.32±.05	92.45±.03	72.22±.10	83.76±.06
		IF-WMV	88.22±.02	96.53±.01	49.09±.04	89.93±.03	75.00±.11	87.52±.05
	Re	EF	79.30±.07	93.65±.03	29.29±.11	82.19±.08	56.94±.17	77.96±.07
		LF	82.01±.05	96.83±.02	36.77±.11	82.37±.08	79.17±.18	88.14±.07
		IF	**92.20±.01**	98.29±.01	**61.17±.07**	**92.81±.02**	87.50±.05	90.13±.01
		IF-WMV	90.61±.04	**98.44±.01**	55.65±.10	90.83±.05	**88.89±.10**	**94.54±.03**

Table 7
Performance of the EXP.1 (Task B1) the COMPLETE dataset. Ba and Re denote the BasicNet and ResNet architectures.

Data	Net	Mod.	ACC	Precision		Recall		AUC
				C	NC	C	NC	
COMPLETE	Ba	EF	80.53±.01	88.64±.02	61.63±.03	84.34±.02	69.94±.03	81.81±.01
		LF	79.72±.01	89.27±.02	59.58±.42	82.32±.06	72.47±.03	81.25±.02
		IF	81.80±.03	90.62±.02	62.94±.04	83.94±.05	75.84±.08	82.77±.03
	Re	EF	82.24±.02	91.58±.15	63.21±.05	83.54±.02	78.65±.04	83.63±.02
		LF	80.83±.02	91.78±.14	60.43±.04	81.21±.01	79.78±.03	83.73±.01
		IF	**86.70±.02**	**93.74±.01**	**71.12±.07**	**87.78±.03**	**83.71±.03**	**87.24±.01**

6.1. EXP.1: Task B1 (C/NC classification)

Tables 4, 5, 6, 7 and Fig. 6 report the performance of the implemented experiments on task B1, for each dataset. On the MRI and PET sets, the proposed Multi Input-Multi Output architecture outperforms the unimodal approach in terms of ACC and AUC, regardless of the involved CNN. Moreover, it exhibits the highest improvement when the ResNet is considered, reporting an increment in AUC of about 2.59% and 5.50% in Tables 4 and 5, respectively.

Table 6 focuses on the PAIRED dataset, where the IF modality achieves at least 90% of accuracy on both networks, surpassing both the EF and LF fusion modalities with an increment of about 7.17% (with regard to EF) and 10.19% (with regard to LF) on BasicNet and ResNet respectively, in comparison with the runner-up solutions. In Table 7, despite the good results achieved by EF and LF, the IF approach remains the top-performing solution.

The bar plots in Fig. 6 highlight that, for each dataset, the proposed architecture reports the best results in terms of BAcc and F1, showing the highest improvements when the ResNet is considered.

Fig. 7 illustrates the training and validation curves region for the experiments involving the IF strategy. Notably, both BasicNet and ResNet architecture demonstrate convergence at around 100 epochs.

6.2. EXP.2: Task B2 (M/S classification)

Tables 8, 9, 10, 11 and Fig. 8 focus on the EXP.2, showing the results on the Task B2. On the MRI dataset (Table 8), the IF modality outperforms the U approach, showing an improvement of about 13.36% and 6.64% on ACC and AUC, respectively, when the ResNet is employed. It also demonstrates the best performance in terms of precision and recall for each class. Similarly, in Table 9, the IF approach achieves an increase in AUC of about 11.50% and 2.92% on BasicNet and ResNet, respectively. Moreover, it also reports an improvement in precision and recall, when compared to the U approach.

On the PAIRED dataset in Table 10, the IF-WMV obtains the best performance for the BasicNet in terms of ACC, Precision, and AUC, while, when the ResNet is considered, IF and IF-WMV report the same values of accuracy, precision, and recall, but differ of about 8.99% on AUC. Moreover, Table 11 focusing on the COMPLETE set, highlights that among all the MDL fusion modalities, the IF introduces an improvement in all the metrics for both networks. Finally, the values of BAcc and F1 presented in the bar plots of Fig. 8, confirm the proposed Multi Input–Multi Output architecture as the best-performing solution. The training and validation curves region illustrated in Fig. 9 describe the convergence of the networks, that is around 50 epochs for both BasicNet and ResNet.

Fig. 6. Performance of the EXP.1 (Task B1) for each dataset in terms of BAcc and F1. Ba and Re denote the BasicNet and ResNet architectures.

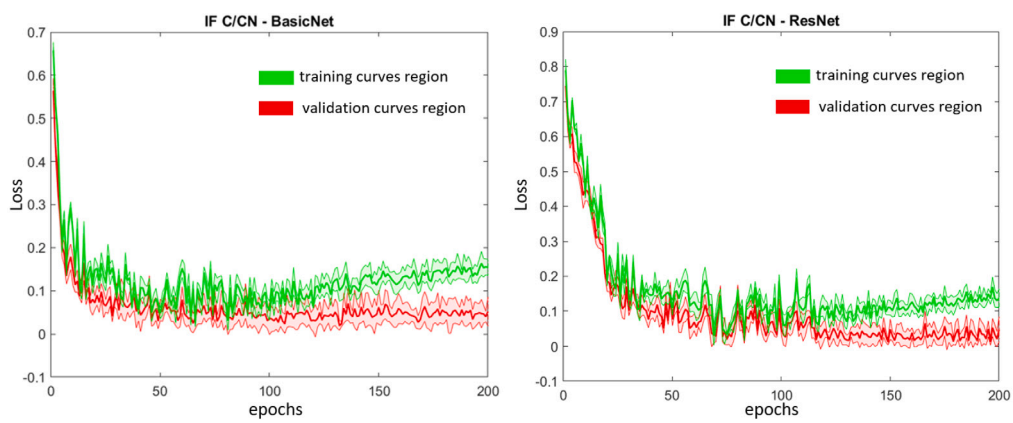


Fig. 7. Training and validation loss curves obtained in the experiments involving the IF approach for Task B1. The lines in bold represent the average training and validation curves computed considering the 5-fold CV, while the underlying area determines the region identified by the standard deviation.

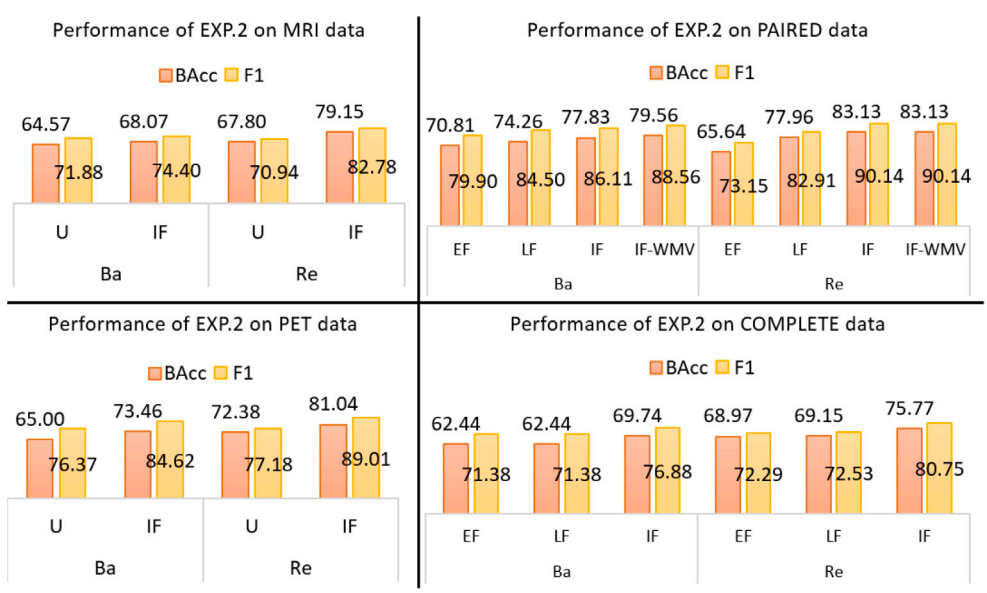


Fig. 8. Performance of the EXP.2 (Task B1) for each dataset in terms of BAcc and F1. Ba and Re denote the BasicNet and ResNet architectures.

Table 8
Performance of the EXP.2 (Task B2) on the MRI dataset. Ba and Re denote the BasicNet and ResNet architectures.

Data	Net	Mod.	ACC	Precision		Recall		AUC
				M	S	M	S	
MRI	Ba	U	70.62±.09	84.12±.05	40.38±.16	75.97±.12	53.16±.14	70.05±.06
		IF	73.29±.03	85.90±.39	44.66±.06	77.91±.06	58.23±.14	70.52±.05
	Re	U	68.84±.08	86.96±.02	40.00±.10	69.77±.10	65.82±.02	68.80±.06
		IF	**82.20±.04**	**91.25±.02**	**59.79±.07**	**84.88±.05**	**73.42±.04**	**75.44±.05**

Table 9
Performance of the EXP.2 (Task B2) on the PET dataset. Ba and Re denote the BasicNet and ResNet architectures.

Data	Net	Mod.	ACC	Precision		Recall		AUC
				M	S	M	S	
PET	Ba	U	74.73±.07	88.24±.08	34.78±.10	80.00±.11	50.00±.29	68.92±.07
		IF	84.62±.05	90.67±.03	56.25±.23	90.67±.06	56.25±.14	80.42±.07
	Re	U	74.73±.03	91.94±.01	37.93±.09	76.00±.03	68.75±.11	77.83±.06
		IF	**89.01±.07**	**93.33±.04**	**68.75±.23**	**93.33±.06**	**68.75±.16**	**80.75±.13**

Table 10
Performance of the EXP.2 (Task B2) on the PARIED dataset. Ba and Re denote the BasicNet and ResNet architectures.

Data	Net	Mod.	ACC	Precision		Recall		AUC
				M	S	M	S	
PAIRED	Ba	EF	79.17±.10	89.09±.04	47.06±.28	84.48±.11	57.14±.09	77.34±.06
		LF	84.72±.09	89.83±.06	61.54±.28	91.38±.11	57.14±.10	84.61±.07
		IF	86.11±.07	91.38±.01	64.29±.28	91.38±.094	64.29±.12	85.71±.12
		IF-WMV	88.89±.05	91.67±.01	75.00±.22	**94.83±.07**	64.29±.11	88.42±.06
	Re	EF	70.83±.14	87.76±.08	34.78±.22	74.14±.19	57.14±.36	71.55±.09
		LF	81.94±.08	92.45±.03	52.63±.09	84.48±.02	**71.43±.11**	84.48±.08
		IF	**90.28±.06**	**93.22±.04**	**76.92±.17**	**94.83±.05**	**71.43±.18**	81.90±.20
		IF-WMV	**90.28±.06**	**93.22±.04**	**76.92±.17**	**94.83±.05**	**71.43±.18**	**90.89±.08**

Table 11
Performance of the EXP.2 (Task B2) on the COMPLETE dataset. Ba and Re denote the BasicNet and ResNet architectures.

Data	Net	Mod.	ACC	Precision		Recall		AUC
				M	S	M	S	
COMPLETE	Ba	EF	70.22±.07	83.40±.04	37.86±.12	76.73±.10	48.15±.13	68.57±.06
		LF	70.22±.07	83.40±.04	37.86±.11	76.73±.09	48.15±.13	68.53±.06
		IF	76.12±.03	86.82±.04	47.96±.10	81.45±.07	58.02±.12	72.26±.07
	Re	EF	70.22±.05	87.89±.02	40.60±.07	71.27±.07	66.67±.02	69.13±.07
		LF	70.51±.06	87.95±.02	40.91±.07	71.64±.07	66.67±.02	69.58±.07
		IF	**83.71±.42**	**89.84±.03**	**55.00±.07**	**83.64±.04**	**67.90±.07**	**77.5±.06**

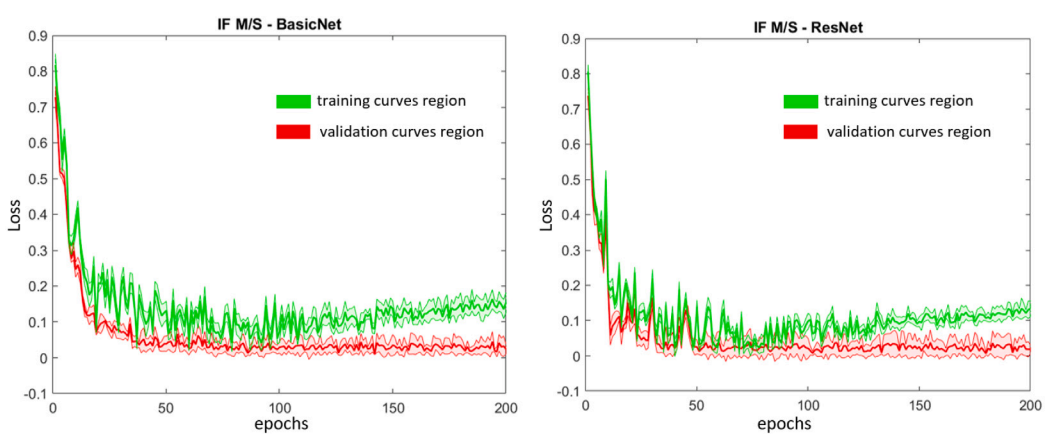


Fig. 9. Training and validation loss curves obtained in the experiments involving the IF approach for Task B2. The lines in bold represent the average training and validation curves computed considering the 5-fold CV, while the underlying area determines the region identified by the standard deviation.

6.3. EXP.3: Task A (C/M/S classification)

The results of the EXP.3 are reported in Tables 12, 13, 14, 15 and Fig. 10 and highlight that the proposed IF modality introduces an improvement on each dataset when compared to the other approaches. Indeed, in Table 12, it outperforms the U approach when both BasicNet and ResNet are involved. Moreover, the IF presents a degradation in performance when the solution proposed in [11] is implemented, reporting a variation of −12.98% and −2.45% in ACC and AUC respectively. Similarly, in Table 13, the use of the IF corresponds to an increase in performance when the proposed architecture is considered, regardless the CNN used as the backbone. Moreover, although the

Table 12
Performance of the EXP.3 (Task A) on the MRI dataset. Ba and Re denote the BasicNet and ResNet architectures.

Data	Net	Mod.	ACC	Precision			Recall			AUC
				C	M	S	C	M	S	
MRI	Ba	U	70.86±.02	86.96±.04	37.42±.07	34.31±.09	80.67±.04	43.80±.02	44.30±.16	76.53±.04
		IF	72.60±.03	87.96±.02	41.37±.04	34.71±.16	**82.07±.06**	44.57±.09	53.16±.13	**79.39±.04**
	Re	U	70.23±.03	91.81±.03	41.34±.05	32.82±.01	73.87±.03	62.02±.03	54.43±.07	78.53±.03
		IF	**75.67±.01**	**91.75±.03**	**49.55±.03**	**38.66±.04**	80.54±.010	**63.57±.04**	**58.23±.05**	78.78±.02
	Re [11]	U	61.36±.04	84.47±.06	27.12±.06	16.33±.07	72.14±.10	32.17±.19	30.38±.02	71.11±.03
		IF	48.38±.05	86.84±.04	26.06±.02	15.17±.06	46.33±.08	59.69±.17	34.18±.22	68.66±.06

Table 13
Performance of the EXP.3 (Task A) on the PET dataset. Ba and Re denote the BasicNet and ResNet architectures.

Data	Net	Mod.	ACC	Precision			Recall			AUC
				C	M	S	C	M	S	
PET	Ba	U	73.14±.04	94.46±.02	21.89±.05	16.22±.39	76.94±.04	49.33±.08	37.50±.07	75.99±.06
		IF	79.32±.04	93.85±.02	26.95±.07	**41.18±.29**	83.71±.05	50.67±.08	43.75±.14	77.73±.05
	Re	U	79.32±.03	**96.76±.01**	33.10±.09	20.45±.09	81.94±.02	62.67±.11	56.25±.03	80.70±.03
		IF	**82.84±.03**	96.35±.01	**38.28±.12**	34.29±.32	**85.16±.03**	**65.33±.07**	**75.00±.23**	**81.80±.05**
	Re [11]	U	65.96±.09	95.53±.02	18.54±.07	8.73±.07	69.68±.08	37.33±.14	56.25±.37	74.76±.05
		IF	76.93±.07	89.13±.03	3.70±.04	6.49±.43	87.26±.09	1.33±.02	31.25±.38	70.34±.05

Table 14
Performance of the EXP.3 (Task A) on the PAIRED dataset. Ba and Re denote the BasicNet and ResNet architectures.

Data	Net	Mod.	ACC	Precision			Recall			AUC
				C	M	S	C	M	S	
PAIRED	Ba	EF	78.82±.05	93.45±.03	18.60±.09	21.05±.18	84.71±.07	27.59±.14	57.14±.37	76.97±.07
		LF	76.11±.04	96.88±.02	24.05±.08	27.27±.45	78.06±.05	65.52±.19	42.86±.35	84.06±.08
		IF	**88.06±.04**	94.85±.14	40.28±.28	**66.67±.06**	92.81±.12	50.00±.09	57.14±.05	86.15±.05
		IF-WMV	87.74±.05	94.99±.01	40.26±.15	**66.67±.33**	**92.09±.06**	53.45±.16	57.14±.09	87.34±.04
	Re	EF	63.54±.26	95.05±.01	15.38±.07	8.16±.08	65.65±.30	51.72±.21	28.57±.21	74.81±.05
		LF	78.98±.07	96.81±.03	28.18±.36	20.83±.13	81.83±.06	53.45±.16	**71.43±.29**	88.84±.08
		IF	**88.06±.01**	97.10±.01	**47.19±.10**	38.10±.19	90.47±.02	**72.41±.08**	57.14±.09	89.45±.04
		IF-WMV	86.78±.02	96.69±.01	43.33±.08	40.00±.33	89.21±.03	67.24±.12	**71.43±.25**	**91.70±.02**
	Re [11]	IF	77.87±.06	93.79±.02	16.67±.14	15.79±.12	84.17±.09	20.69±.31	64.29±.25	77.10±.10
		IF-WMV	78.03±.06	93.60±.02	17.11±.12	17.31±.13	84.17±.09	22.41±.29	64.29±.25	76.98±.08

Table 15
Performance of the EXP.3 (Task A) on the COMPLETE dataset. Ba and Re denote the BasicNet and ResNet architectures.

Data	Net	Mod.	ACC	Precision			Recall			AUC
				C	M	S	C	M	S	
COMPLETE	Ba	EF	73.70±.03	87.54±.03	41.78±.06	37.89±.13	83.74±.03	46.18±.14	44.44±.16	77.02±.03
		LF	70.43±.03	87.56±.03	37.39±.06	34.65±.12	78.89±.03	48.00±.12	43.21±.12	76.36±.03
		IF	76.89±.03	87.93±.02	47.73±.09	43.75±.14	**87.58±.05**	45.82±.08	51.85±.11	80.53±.02
	Re	EF	72.88±.01	91.73±.02	43.55±.02	33.59±.10	78.38±.02	58.91±.03	53.09±.07	81.19±.01
		LF	72.66±.01	**91.90±.02**	44.20±.02	31.94±.06	77.98±.05	58.18±.02	56.79±.06	81.43±.01
		IF	**77.93±.02**	91.80±.02	**51.35±.06**	**45.05±.12**	83.64±.28	**62.18±.03**	**61.73±.04**	**82.81±.01**
	Re [11]	IF	62.70±.04	87.13±.02	33.76±.03	16.47±.06	69.09±.07	48.00±.16	34.57±.21	76.98±.04

solution described in [11] presents an improvement in the ACC in comparison with the U approach, it obtains lower values in AUC, Precision for each class, and Recall for the M and S levels.

On the PAIRED dataset in Table 14, the IF represents the best-performing solution, surpassing by a wide margin the other methods. Moreover, acceptable performance can also be seen in the results obtained by the state-of-the-art methodology described in [11]. In Table 15, it is possible to note that although EF and LF present good performance, the IF modality introduces an improvement in terms of ACC and AUC for BasicNet and ResNet respectively. Finally, the values of BAcc and F1 presented in the bar plots of Fig. 10, confirm the proposed Multi Input-Multi Output architecture as the best-performing solution.

Fig. 11 shows the training and validation loss curves obtained in the Task A involving the IF strategy. It is possible to note, that both BasicNet and ResNet demonstrate convergence at around 100 epochs.

6.4. EXP.4: Task B (C/M/S hierarchical classification)

Tables 16, 17, 18, 19 and Fig. 12 focus on EXP.4, showing the results on Task B. On MRI and PET datasets (Tables 16 and 17), the use of the IF approach corresponds to an improvement of the performance in comparison with the U modality, regardless of the involved CNN. Moreover, the greatest increase is obtained when the ResNet is involved.

On the PAIRED dataset in Table 18, the IF is confirmed as the best-performing solution, surpassing by a wide margin the other modalities.

Fig. 10. Performance of the EXP.3 (Task A) for each dataset in terms of BAcc and F1. Ba and Re denote the BasicNet and ResNet architectures.

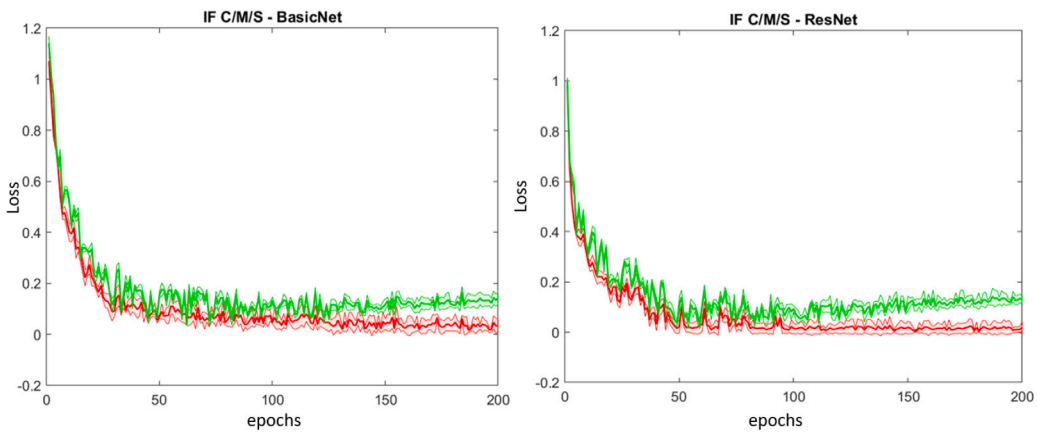


Fig. 11. Training and validation loss curves obtained in the experiments involving the IF approach for Task A. The lines in bold represent the average training and validation curves computed considering the 5-fold CV, while the underlying area determines the region identified by the standard deviation.

Table 16
Performance of the EXP.4 (Task B) the MRI dataset. Ba and Re denote the BasicNet and ResNet architectures.

Data	Net	Mod.	ACC	Precision			Recall			AUC
				C	M	S	C	M	S	
MRI	Ba	U	70.70±.01	89.54±.01	46.91±.03	24.32±.18	77.65±.02	50.00±.07	56.96±.16	77.92±.01
		IF	72.53±.03	89.35±.02	42.62±.03	31.97±.15	80.67±.05	50.39±.09	49.37±.13	80.51±.03
	Re	U	73.00±.03	91.96±.01	46.26±.05	28.16±.05	80.24±.02	50.39±.09	62.03±.06	78.93±.03
		IF	**80.29±.01**	**93.28±.02**	**55.08±.03**	**50.00±.04**	**85.42±.01**	**65.12±.03**	**69.62±.05**	**82.00±.02**

Table 17
Performance of the EXP.4 (Task B) the PET dataset. Ba and Re denote the BasicNet and ResNet architectures.

Data	Net	Mod.	ACC	Precision			Recall			AUC
				C	M	S	C	M	S	
PET	Ba	U	76.09±.04	94.51±.02	26.87±.04	12.24±.39	80.48±.04	48.00±.08	37.50±.07	76.87±.04
		IF	78.62±.03	96.34±.02	28.81±.06	53.33±.28	80.65±.05	68.00±.08	50.00±.13	83.49±.04
	Re	U	85.51±.03	95.90±.01	43.37±.09	25.58±.08	90.48±.02	48.00±.10	**68.75±.06**	78.58±.03
		IF	**88.33±.03**	**96.91±.01**	**47.32±.12**	**64.71±.29**	**90.97±.02**	**70.67±.06**	**68.75±.23**	**84.13±.05**

Indeed, the maximum increase of ACC is achieved with regard to the EF in both networks, reporting an improvement of about 10.99% and 14.49% for BasicNet and ResNet, respectively. On the COMPLETE dataset, it is possible to note that the IF modality outperforms both EF and LF in all metrics. Finally, Fig. 12 shows the performance in terms of BAcc and F1, confirming the proposed Multi Input-Multi Output architecture as the best solution and the supremacy of the ResNet in comparison to the BasicNet. It is worth noting that in EXP.4 we apply

Table 18
Performance of the EXP.4 (Task B) the PAIRED dataset. Ba and Re denote the BasicNet and ResNet architectures.

Data	Net	Mod.	ACC	Precision			Recall			AUC
				C	M	S	C	M	S	
PAIRED	Ba	EF	81.21±.04	93.42±.03	21.43±.08	46.15±.43	86.87±.05	36.21±.18	42.86±.25	79.42±.08
		LF	77.87±.03	96.72±.02	26.57±.07	29.63±.47	79.68±.05	65.52±.19	57.14±.38	87.21±.08
		IF	88.85±.04	96.25±.03	**57.38±.09**	27.27±.05	92.45±.11	60.34±.09	64.29±.04	84.67±.04
		IF-WMV	88.85±.04	96.60±.01	55.88±.13	29.03±.29	91.91±.05	65.52±.11	64.29±.10	89.23±.04
	Re	EF	76.59±.07	93.65±.02	24.62±.38	10.67±.15	82.19±.07	27.59±.19	57.14±.36	78.31±.08
		LF	80.25±.07	96.65±.02	29.46±.34	25.64±.15	82.91±.04	56.90±.15	**71.43±.29**	88.58±.07
		IF	**91.08±.01**	98.29±.01	53.33±.09	61.54±.16	**92.81±.02**	**82.76±.10**	57.14±.09	88.73±.04
		IF-WMV	89.01±.01	**98.43±.01**	45.71±.09	**69.23±.24**	90.29±.02	**82.76±.16**	64.29±.20	**94.38±.01**

Table 19
Performance of the EXP.4 (Task B) the COMPLETE dataset. Ba and Re denote the BasicNet and ResNet architectures.

Data	Net	Mod.	ACC	Precision			Recall			AUC
				C	M	S	C	M	S	
COMPLETE	Ba	EF	74.37±.03	88.90±.03	44.08±.04	32.38±.12	84.14±.03	48.73±.11	41.98±.15	79.52±.03
		LF	71.99±.02	89.14±.03	40.71±.04	30.70±.11	80.40±.03	50.18±.11	43.21±.11	78.73±.02
		IF	76.97±.02	90.73±.01	54.81±.07	31.08±.11	85.05±.04	53.82±.07	56.79±.10	82.81±.02
	Re	EF	75.71±.01	91.66±.02	50.19±.02	30.00±.08	84.34±.02	48.36±.03	62.96±.07	80.88±.02
		LF	73.92±.02	91.70±.03	45.82±.03	30.54±.06	81.52±.03	49.82±.02	62.96±.05	80.96±.01
		IF	**82.32±.02**	**93.72±.01**	**58.07±.04**	**55.00±.11**	**87.47±.20**	**68.00±.02**	**67.90±.03**	**84.97±.01**

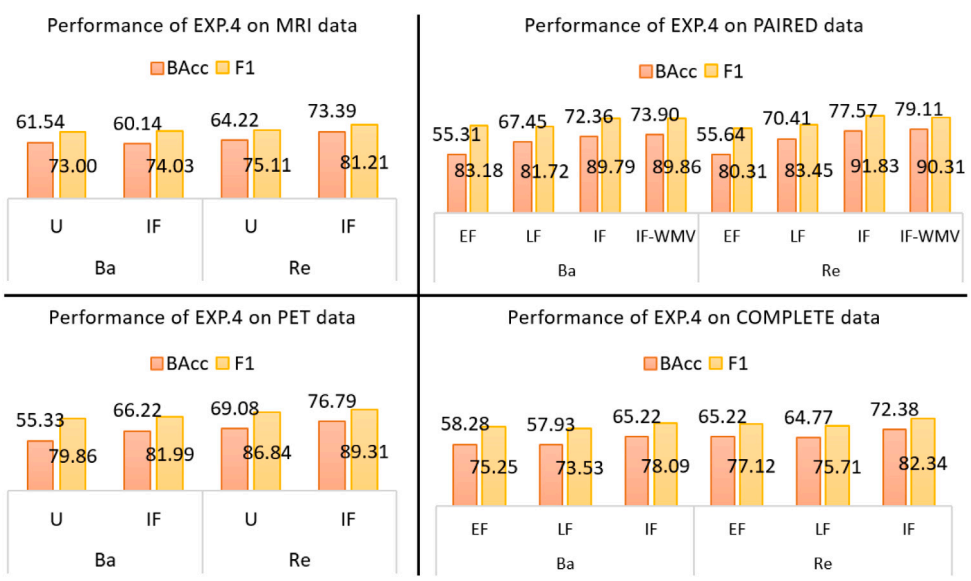


Fig. 12. Performance of the EXP.4 (Task B) for each dataset in terms of BAcc and F1. Ba and Re denote the BasicNet and ResNet architectures.

a hierarchical classification exploiting the prediction of the tasks B1 and B2. As a consequence, we do not report the training and validation curves.

7. Discussion and conclusions

Results show that for each experiment and for both investigated networks (BasicNet and ResNet), the solutions that involve the proposed Multi Input-Multi Output 3D CNN have the best performance, even when the three sub-networks are evaluated separately. Moreover, ResNet outperforms BasicNet in each experiment, confirming, as expected, the effectiveness of fine-tuning in the medical field. In contrast to BasicNet, ResNet can exploit the knowledge learnt during the previous training phase.

The analysis conducted with the PAIRED and COMPLETE datasets in Section 6 shows that in most experiments, the LF performed better than the EF, thus supporting the preference of the former over the latter. Indeed, the LF approach exploits two different networks, each trained for a specific image modality, leveraging a set of data larger than the one used in the EF. As a consequence, the two CNNs learn to extract features that reflect the distinctive characteristics of each modality, delaying the combination of the results in a post-processing step. On the other hand, in the EF solution, that exploits only the PAIRED dataset in the training phase, the MRI-PET pairs simultaneously contribute to the prediction, as they are organized in a series of 2-channel 3D volumes. However, we argue that the limited amount of data and the structure of the input do not allow the network to learn a shared features representation that at the same time is able to capture the distinctive characteristics of MRI and PET. As a consequence, in the case of a highly unbalanced and incomplete dataset, focusing only on the paired acquisitions may represent a limitation for the application of multimodal approaches, resulting in the need to introduce an efficient training strategy that leverages all the available volumes. In solutions involving the proposed IF approach, the shared representation is created by concatenating features coming from the convolutional cores at an intermediate level, thus preserving the distinctiveness of the different image modalities, which is then exploited in the classification core in the case of MRI-PET pair. However, in the implemented training strategy, the loss is propagated back to both the MRI and PET convolutional cores when a *paired acquisition* is considered as input, leading the two features

Table 20
Statistical significance of the proposed Multi Input–Multi Output 3D CNN. The IF is compared with the solutions involving the U (IF vs U), EF (IF vs EF), and LF (IF vs LF) modalities using a paired t-test with a significance level of 0.05 that considers the probability distributions of predictions from different networks as paired observations.

Task	Net	IF vs U		IF vs EF		IF vs LF	
		MRI	PET	PAIRED	COMPLETE	PAIRED	COMPLETE
B1	Ba	$3.21 \cdot 10^{-11}$	$\mathbf{9.06} \cdot 10^{-2}$	$5.12 \cdot 10^{-16}$	$9.41 \cdot 10^{-8}$	$3.56 \cdot 10^{-6}$	$2.51 \cdot 10^{-2}$
	Re	$1.26 \cdot 10^{-12}$	$2.30 \cdot 10^{-3}$	$1.04 \cdot 10^{-4}$	$4.22 \cdot 10^{-15}$	$1.34 \cdot 10^{-23}$	$8.77 \cdot 10^{-18}$
B2	Ba	$\mathbf{3.72} \cdot 10^{-1}$	$6.63 \cdot 10^{-5}$	$\mathbf{8.91} \cdot 10^{-1}$	$\mathbf{7.48} \cdot 10^{-2}$	$6.10 \cdot 10^{-3}$	$3.20 \cdot 10^{-2}$
	Re	$1.07 \cdot 10^{-7}$	$1.15 \cdot 10^{-5}$	$5.30 \cdot 10^{-3}$	$1.25 \cdot 10^{-10}$	$5.09 \cdot 10^{-7}$	$5.82 \cdot 10^{-11}$
A	Ba	$\mathbf{6.53} \cdot 10^{-1}$	$1.39 \cdot 10^{-12}$	$2.78 \cdot 10^{-8}$	$6.61 \cdot 10^{-5}$	$7.96 \cdot 10^{-16}$	$1.68 \cdot 10^{-4}$
	Re	$7.60 \cdot 10^{-3}$	$7.24 \cdot 10^{-4}$	$2.15 \cdot 10^{-4}$	$2.47 \cdot 10^{-6}$	$2.54 \cdot 10^{-8}$	$3.47 \cdot 10^{-6}$

extracting processes to create a shared representation that is suitable for the task to solve. Moreover, the proposed Multi Input–Multi Output 3D CNN, consisting of three sub-networks where the PAIRED-NET exploits the convolutional cores of MRI-NET and PET-NET (Fig. 3(c)), allows the intermediate fusion method to use all the data available in the dataset and classify also the instances where one modality is missing. As a result, the IF approach is able to overcome the disadvantages identified in the EF and LF solutions, introducing an improvement in the performance obtained and, thus, representing the best solution among all those described and implemented in this paper. To assess the statistical significance of the proposed Multi Input–Multi Output 3D CNN compared to other approaches, we conduct a paired t-test with a significance level of 0.05. This analysis considers the probability distributions of predictions from different networks as paired observations, providing a robust evaluation of the model's superiority. In particular, in Table 20, we compare the IF with the solutions involving the U (IF vs U), EF (IF vs EF), and LF (IF vs LF) modalities, reporting the p-value of the tests for each dataset. We detail the statistical analysis for each task, considering both the BasicNet and ResNet. Notably, the proposed IF approach consistently exhibits a significant difference from other solutions (p-value < 0.05) when the ResNet is involved, affirming the effectiveness of fine-tuning in the medical domain. Indeed, the results of the comparison reported in the case of BasicNet indicate that in a few experiments, the statistical significance of the proposed Multi Input–Multi Output 3D CNN cannot be confirmed (p-value > 0.05).

When comparing Tables in in Sections 6.3 and 6.4 the hierarchical classification (Task B) has better performance than the three-class classification (Task A). We deem that this happens because in Task B we explicitly exploit the existing relationship between classes, while in Task A we delegate the model the role of understanding the connection between the labels. However, the small size and high imbalance of the dataset make this task very complex.

We also consider the work proposed in [11] representing the first article considering MDL approaches for the OASIS-3 dataset [15]. Tables 12, 13, 14 and 15 show that our methodology outperforms the solution proposed in [11] in all metrics and for each dataset. Our approach differs from that presented in [11] for the characteristic of the CNN involved and for the strategy implemented to handle the incomplete dataset, consisting of replacing missing modalities with black images. We argue that these characteristics negatively affect the performance. Furthermore, the 2D CNN implemented in [11] only considers the median slice as input without exploiting the volumetric information of the involved acquisitions, while the use of black images may affect the process of features learning, especially if the number of incomplete inputs is greater than the number of paired acquisitions. As a consequence, the solution described in this paper takes these aspects into account by proposing a training strategy that handles both paired and incomplete data. When comparing our approach with other proposals in the literature, we note that the implemented methodology has performance comparable to the solution proposed in [24] for EXP.1 and EXP.3 on the MRI dataset. The authors in [24] proposed a 3D CNN trained with MRI volumes and reported only the accuracy and the balanced accuracy to assess the performance. In particular, in the case of Tasks B1 and A the authors achieved values of accuracy equal to 84% and 77% respectively; moreover, in the same tasks the balanced accuracy is 84% and 76%, respectively. We do not compare our proposal with the other state-of-art solutions focusing on the OASIS-3 dataset [15,18,19,21,23,25,38] since they conducted an image-level rather that a patient-level split of the dataset, introducing a bias in the performance evaluation step.

To the best of our knowledge, our methodology represents the first work to provide an analysis of different MDL fusion techniques for dementia severity assessment using OASIS-3 dataset [15], completely relying on 3D CNN, and implementing in the IF approach a network architecture that automatically handles an incomplete dataset both during the training and inference step.

Despite the promising results, we argue that the limitations of our work are the limited size of the dataset and the quality of acquired medical images. Indeed, as mentioned in Section 3 and in the *Supplementary Material*, we implemented a pre-processing step that excluded several patients reducing the number of subjects from 1098 to 628. In addition, the OASIS-3 dataset [15], consists of a collection of different studies over the course of 15 years, including MRI and PET acquired with different scanners and protocols, that may limit the generalization ability of our models. To address this, we evaluate the difference in the performance among the various machines, detailing the results for each MRI and PET scanner. In particular, in the *Supplementary Material* we provide all the values for each classification task in different tables, reporting the results obtained with the unimodal approach and the proposed IF strategy. It is worth noting that although we do not implement any advanced data harmonization techniques, the performance does not vary significantly among different scanners. Moreover, the MRI and PET machines belong to the same manufacturer, that is the SIEMENS, while different producers should be included to provide a more thorough evaluation of the generalization ability of our models on various devices.

Future works will continue to explore the intermediate fusion modality, further analyzing the properties of the shared features representation. In particular, we will focus on the study of approaches that aim to improve the integration of heterogeneous data, generalizing them to case studies with more than two modalities. Moreover, we will work on the explainability of the implemented models, evaluating the decisions made by the networks in comparison with the clinical diagnosis.

CRediT authorship contribution statement

Michela Gravina: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Angel García-Pedrero:** Funding acquisition, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing. **Consuelo Gonzalo-Martín:** Funding acquisition, Project administration, Resources, Supervision,

Writing – original draft, Writing – review & editing, Validation. **Carlo Sansone:** Funding acquisition, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing. **Paolo Soda:** Funding acquisition, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.artmed.2024.102774.

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